OASIS + The GOAT

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Solutions to problems in ORAC. OASIS is very summarised and could be used as hint-ish editorials. However, The GOAT contains more useful general tricks from past problems.

The GOAT (AIO Problems)

The Good Orac Answers Table (And Interesting Other Problems)

"It would take too long to summarise every problem, right?"

A bit more detailed than OASIS, only actual good techniques/observations are here. Also includes some random cf, atcoder, and nzic problems that I like. Right now, no content, so scroll down

Task ID	**General Idea**	**Tags**
simplerect	?	?
oracdatastacksbracket	?	?
pre09primes	?	?
dsinversioncounting	?	?
mockaio26sum	?	?
mockaio26atlantis	?	?
aio25track	?	?
aio25cards	?	?
aio24backpacking	?	?
aio24tennis	?	?
aio23shop	?	?
aio22composing	?	?
aio21space	?	?
aio21space	?	?
aio20tennis	?	?
aio19evading	?	?
aio19lsc	?	?
aio18janitor	?	?
aio17lds	?	?
aio16probe	?	?
aio16balance	?	?
aio15ruckus	?	?
aio14garden	very similar to 14chimera	?
aio13art	?	?
aio10snap	?	?
aio08surfing	?	?
aio07invasionsen	?	?
aio06mouse	?	?
aio06atlantis	?	?
alpha22hippos	?	?
aiio26cherry	?	?
aiio25buildings	?	?
aiio25retail	?	?
aiio24friend	?	?
aiio23fault	?	?
aiio22winter	?	?
aiio20metro	?	?

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Task ID	General Idea	Tags
aiio13tower	?	?
aiio05spies	?	?
fario10digicode	?	?
fario10abc	?	?

OASIS

ORAC Answers/Solutions In Summary

"It would be useful to have the solution to every AIO problem I've solved"

Task ID	Main Idea	Tags
simpleadd	Add 2 numbers	simple
simplesit	Maths, casework	simple
simplemixed	Division and remainders	simple
simpletaktak	Modulo	simple
simplecount	Implement for-loop	simple
simplestats	Find mode, median, range	simple
simpledrought	USe for-loop	simple
simplelookup	Dictionary	simple
simplesearch	Dictionary	simple
simpletrip	Modulo 3	simple
simplefriend	Dictionary	simple
simplepal	Palindrome Checking	simple
simplerect	Rectangle Intersection	simple
oracsortbernard	Implement a binary search algorithm	bsearch
oracsortnotso	Sort an array	sorting
oracsortmode	Find the mode	simple
oracsortswaps	Sort with swaps	loopn
oracstacks3	Implement a stack	basicds
oracqueues2	Implement a queue	basicds
oracdatastacksbracket	Sweep through to find if bracket sequence is valid	loopn
pre09primes	Implement sieve of erastosthenes	math
pre09pairs	Simultaneous loops at different points	loopn
oracgraphadjlist	Read a graph adj list	graph
trial05qsort	Check if all nodes are reachable	dfs/bfs
trial01gossip	Check if nodes are connected	bfs/dfs
oractiling*	Plug DP :(	dp
oracoddjobs*	Complete search with a bitset	bitset
dscannons	Make a dp which relies on previous values, but use monotone queue/segtree to keep track of prev in total	dp, rmq
dsfenwickxor	Implement a fenwick tree with XOR operation	rmq
dsinversioncounting	Pretty much also a dp but optimise with rmq	dp, rmq
dslazyupdate	Implement a lazy segment tree	rmq
dsrangetreeupdates	Implemnt a min segtree	rmq
dssupplies	Implement a range-update, point-check segtree	rmq
aic04genius	Use lots of maths by hand pretty much	math
aic03culture	What is log2(n)	math

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Task ID	Main Idea	Tags
aic99hail	Make a recursive algorithm run n times	recursion
mockaio26center	Maximum median and mean are obtained by highest number in array	obs
mockaio26mode	Find the range where all occurences of x are	ranges
mockaio26sum	Kadane's algorithm but better	dp
mockaio26interception	Use bfs to get distance	dfs/bfs
mockaio26queue	Very simple dp, but track previous states together with pmin	dp
mockaio26atlantis	Double prefix sums, then see if there are (1, -1, -1, 1) tuples by two pointers	psums, 2ptrs, obs
aio25soccer	Keep track of 2 ++ variables	loopn
aio25treasure	Find the intersection of n ranges	loopn
aio25track	Fix a point, then do greedy in O(1)	greedy, fix
aio25orac	Very observation, by subtasks	greedy
aio25cards	Use Extremal principle 2 times, then use set to run greedy	obs, set
aio24javelin	Find # of max-es in array	loopn
aio24subbookkeeper	Find longest consecutive letter, but allowed to insert 1 greedily	loopn
aio24shopping	Classic atcoder problem, easy obs	max
aio24backpacking	From cases to case, dp with precomputation backwards, need to prove exchance argument	greedy, dp
aio24tennis	Realise that there is a net change for each thingy and a min value, check for infinite loop or maths	obs
aio23teletrip	stimulation	stim
aio23raffle	Make freq table, then loop through it	freq
aio23bank	Trivial dp backwards	dp
aio23shop	Make prefix/suffix min and max and sum, then trivial	psum
aio23deal	Binary search and greedy I'm pretty sure	bsearch, greedy
aio22election	Keep track of 3 variables	simple
aio22ground	FInd longest subsequence of same number	loopn
aio22tsp	Find minimum and maximum range possible, intersection but not needed	greedy
aio22buildings	Sweep across array 2 times, and greedy for second time after precomp	loopn
aio22composing	DP both ways increasing and decreasing then get max of mins for all nodes/array index	dp
aio21robot	Stimulate coordinate moving	stim
aio21artclass	Max and min of r and c suffices to cover all	obs
aio21melody	Get frequency of all nums per mod3 index, then maximum	obs
aio21distancing	Extremely easy loopn greedy	greedy
aio21space	Set binary search and array	obs
aio21laser	Keep track of top, bottom, left, right and then easy	obs
aio20baubles	Greedy casework - be very careful	greedy, obs,case
aio20cookies	DP for days or casework greedy	greedy, obs

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Task ID	Main Idea	Tags
aio20ghost	Make array indexes, hten sliding window from starting points, or greedy thingy	coord, slide
aio20tennis	sort heights, then find min that fills all remaining buckets and when it doesn't work (or too much), do last bit	obs
aio20ladybugs	Sliding window in worst case N^2, but can vectorise it	vectorise
aio19vases	x 1 1	obs
aio19rps	Maximise wins, then draws, then rest are lose	obs
aio19hire	Assign max value to max threshold, assign min value to min threshold	greedy
aio19snake	Binary search simple, but can add 2nd one	bsearch
aio19evading	Parity BFS on all nodes modulo 2	dfs/bfs
aio19lsc	?	?
aio18street	Floor division, park blocks of size 1	obs, math
aio18calvary	Can group up knights as a freq table and check divisibility	obs
aio18cloud	Do a sliding window (kinda) of n people and check distance between last and first	slidingwindow
aio18janitor	Stimulat the first round, then for each update check how many squares rely on it (changes)	obs, stim
aio17mango	See that for each person, there are only 2 possible spots. Then match	obs
aio17tag	Simulate, keep track of what team each are in	stim
aio17coconut	If the euclidean distance is $\leq$ sum of radii and more than difference of distances	math
aio17snake	?	?
aio17chimera	Check longest range from point 0, then from next point and so on	loopn
aio17lds	Notice that each statement of $dist[a] \geq dist[b]$ gives a possible range	obs
aio16magic	Annoying casework and some math to do	math, case
aio16farmer	Inwards 2 pointers to the middle, then check if equal	2ptrs
aio16sculpture	[should be better solution] Check max of where to drop	loopn, rmq
aio16probe	Manhattan distance, BUT there is euclidean distance edge case for directly diagonal	obs
aio16balance	Inwards 2 pointers and see if can satisfy otherwise stack	2ptrs, greedy
aio15snap	There is always winner. See manhattan distance mod2	obs
aio15chairs	Either implement a 2ptr algo, or psum # of wet	2ptrs, psum
aio15ruckus	Use dfs to find lines/circles of kids, then greedy split	dfs/bfs
aio15binary	?	?
aio14garden	See that to save a 'range', you must use some thingies, then greedily assign	greedy, obs

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Task ID	Main Idea	Tags
aio14party	Initially invite everyone, delete everyone who knows not enough, then do same for know too many. Updates asw	greedy
aio14chimera	Casework greedy, pretty much try use as much 'potential' that one has	greedy
aio13archery	To maximise, what is the maximum score for both criteria, similar for min.	obs
aio13frog	Use a sparse table (or prefixes), for each post find min after and max between, and other way	?
aio13saveit	Only combine ones that actually help, with mod5	?
aio13art	Rearranging, we get values of max/min r-c/c-r, and then try all	math, graph
aio12nort	O(1), realise if at least 1 coord is even, can do all, otherwise 1 less	obs
aio12posters	See the best as you sweep, needs some intiution	obs
aio12cabinet	?	?
aio12arthur	?	dfs/bfs
aio12frog	Realise this is the median trick for movement, then calculate movement	obs
aio11pirates	Simple casework of which way you go around	case
aio11magic	Stimulate where the ball is	stimulate
aio11aliens	Pretty much sliding window sum and max	slidingwindow
aio11paradise	Keep a set of ranges, and bsearch to find where they cut (observe no ranges have intersection)	obs
aio10ninjas	Number caught is ceildiv	math
aio10choreo	Stimulate the routine and check for needed hoops	loopn
aio10oil	Even though k is pretty big, realistically you only go through every node at least once. Then bfs	obs, dfs/bfs
aio10heat	Check all elements compared to threshold	loopn
aio10snap	Get a frequency table of occurences for each, then multiply combi	math
aio09sales	Take 2 maximal elements and sum	math
aio09nom	Stimulate process, keep track of last one	loopn
aio09safe	Sort array, do sliding window and check equality	slidingwindow
aio09treasure	Check how many neighbours when dfsing	dfs/bfs
aio09carmen	?	?
aio08ladybugs	Get max and min idx then distance	math
aio08atlantis	Get prefix and suffix maxes, then loop through and find min of pref and suff	psums
aio08debts	[Better solution] Get starting point and then rmq on psums to see if negatives	rmq
aio08surfing	Reverse the order of arrows and dfs to find bad points	dfs/bfs
aio08chocolates	Stimulate but take mod 10 and then write eqs	math
aio08dolls	?	?

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Task ID	Main Idea	Tags
aio07wetlands	Stimulate change for every day	loopn
aio07superphone	Extremely simple dag dp	dp
aio07mansionsen	Maximum of all sliding windows	slidingwindow
aio07invasionsen	For each edge, check both sides and have adj set for all countries	obs
aio06fashion	Realise that greedy is enough and just do. However, dp is possible since only 4 transitions	greedy
aio06ants	Keep track of previous, and res	loopn
aio06mouse	See bounding rectangle, then decide which other edge points there are	obs
aio06jujitsu	Use min possible to win/draw, otherwise sack	greedy
aio06spies	Somewhat of a 2 pointers, check intersection until null	2ptrs
aio06atlantis	Simple DP with sortings from high to low	dp
aio05cute	Loop backwards and get # of 0s	loopn
aio05curry	?	greedy
aio05landmark	?	monotonestk
aio05chimpanzee	You can either stimulate the movement, or use maths	math
alpha24bars	Keep a frequency table, and when u add/remove, check net change	loopn
alpha24winter	Build a graph first by sorting by coord type, then run dijkstra's	sort, graph
alpha23chocolate	Keep track of left and right sum, then when move fixed point, O(1)	loopn
alpha23ecofriendly	Get square and rectangle psums, then run dp	dp, psums
alpha23sewers	Build graph with given nodes, then BFS with memory	dfs/bfs
alpha22ramp	Very simple dp kinda like composing pyramids	dp
alpha22jumps	DP but realise there ae only 2 sums that you keep track of, so just assign to var	dp, obs
alpha22hippos	Nice trick, dfs from node 1, then dfs reverse from node n, and only nodes both reachable are	dfs, obs
aiio26cherry	Get 2 sorted sets by 1st and 2nd key, then suffix max. Then for all thingies, binary search their value in both sets and check for smax.	?
aiio25buildings	Get a frequency array, then loop and check if there exists something in the frequency array with desired value. Then, remove current element and move on	freq, loopn
aiio25retail	Do a DP on max number of money achievable with an amount of money, then stimulate days	dp
aiio24friend	Realise that the final friend is either the start or end, and then because you have to get prefix sum AND prefix/suffix min/max, use 2 segtrees, and see which side has a more of a 'block', then ez	obs
aiio23fault	Multi-source dijkstra, subtask is BFS multi	graphs

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Task ID	Main Idea	Tags
aiio22winter	Its a tree! Then just do postorder dp on max subtree thingy	tree
aiio20metro	The hypergrid condition can just be changed to adding a directed edges from each node to the hypergrid with weight c_i, then a directed edge to every other node 0, then it's easy dijkstra	grid, obs
aiio16greece	?	?
aiio13tower	First, even though the tower needs to be technically sorted, there is no order of specific placing, so we can ignore that. Then, its just 0/1 knapsack but each time there are 2 choices instead of 1	dp
aiio12timewarp	Quite tight, but the idea is to run sliding window sum max for each time warper, then sum up rest	slidingwindow
aiio12negotiations	?	bsearch
aiio11bus	First, find the round(s) where each participated and ended, then the greedy is just always see someone at the last match they play, because if you were to see 2 at the same time, it will always occur then	graph, greedy
aiio11kiwi	Find cycles for every single number in $O(N)$, if there are some components not, then impossible. Then lcm of sizes	graph
aiio10nom	Build psums. Then if B takes first n pavolvas, check if psums(2n) = 2psums(n), and find max	psums
aiio09space	For each block, get min and max x-val, then union intervals, then complement of U	?
aiio07snurgle	?	?
aiio05spies	Median trick for x-val and y-vals	obs
fario12chef	Just do the contribution technique thingy and rearrange some equations for O(N)	math
fario11bookshelf	?	?
fario10digicode	Run sort of a recursive dp, with first an adjlist for keys, then use something like prevpossible, and currpossible	dp
fario10abc	Simple string-hashing excercise	strings
fario09bookshop	See the earliest position for each book for O(N)	loopn
fario08graffiti	Somewhat of an annoying casework dp, where you have to get square sums and tricky (sort of) transition	dp
fario04stars	First, build sparse table. Then for each pair of dots, see the max height between and calculate	rmq
seln12career	?	?
seln04jogging	?	?
seln03conveyor	?	?
seln03lab	?	?
seln03monsters	?	?
seln03taxis	?	?

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Task ID	**Main Idea**	**Tags**
seln03choc	?	?
seln03social	?	?
seln99pow2	?	?
seln99night	?	?
betatrial13zigzag	?	?